ON BOARD TRAINING RECORD BOOK

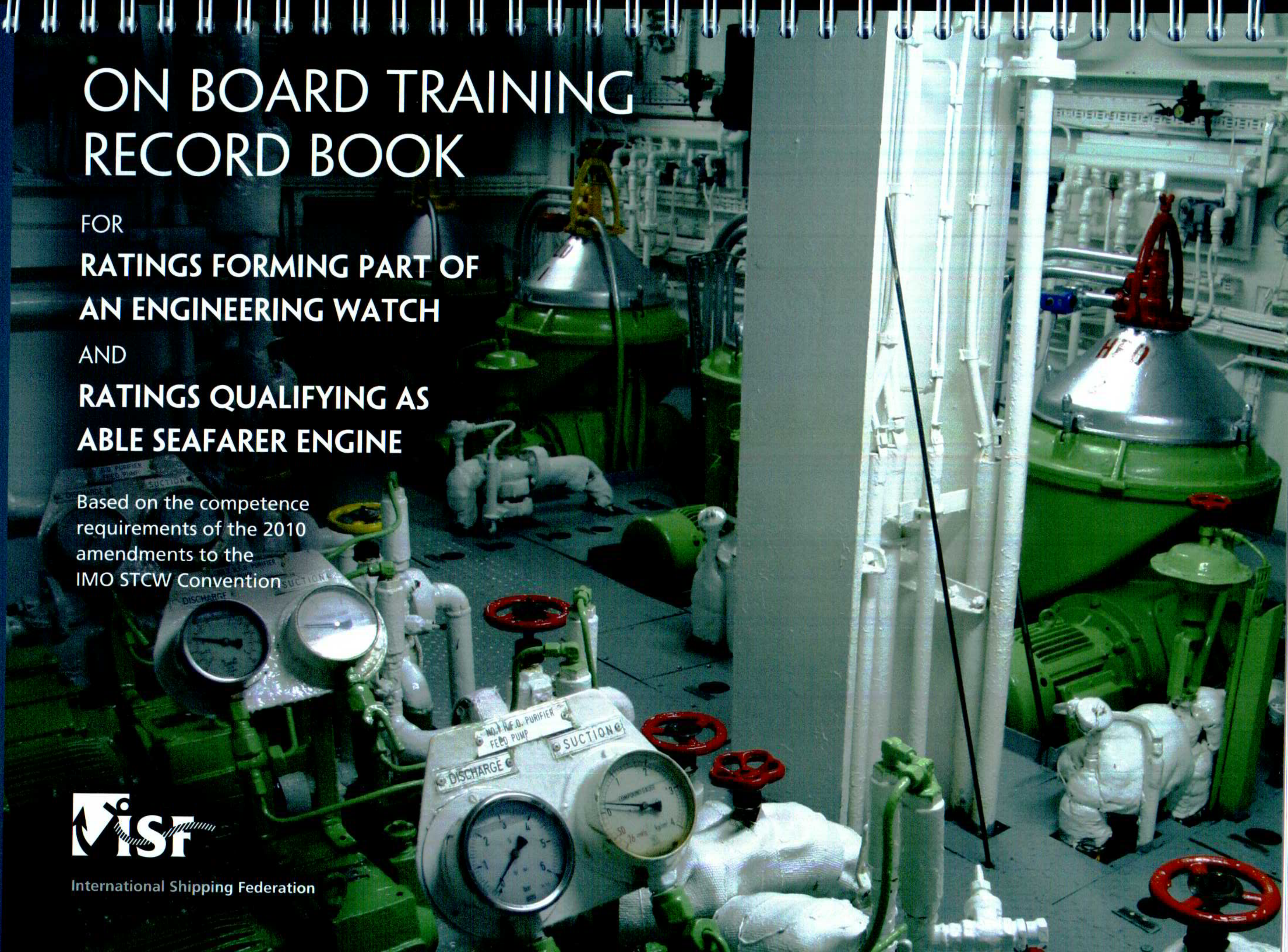
FOR

RATINGS FORMING PART OF AN ENGINEERING WATCH

AND

RATINGS QUALIFYING AS ABLE SEAFARER ENGINE

Based on the competence requirements of the 2010 amendments to the IMO STCW Convention



ISF
International Shipping Federation

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Based on the competence requirements of the 2010 amendments to the IMO STCW Convention

Name ....

Home Address ....

....

....

....

Date Training Started ....

International Shipping Federation

ON BOARD TRAINING RECORD BOOK FOR RATINGS FORMING PART OF AN ENGINEERING WATCH AND RATINGS QUALIFYING AS ABLE SEAFARER ENGINE

SECOND EDITION

Established in 1909, the International Shipping Federation (ISF) is the name used by the International Chamber of Shipping (ICS) when representing maritime employers globally on labour affairs, manpower and training issues. ISF/ICS membership comprises national shipowners' associations from 36 countries. ISF enjoys consultative status with the International Maritime Organization (IMO) and led the representation of maritime employers throughout the revision of the STCW Convention culminating with the Diplomatic Conference, in Manila, in June 2010.

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ON BOARD TRAINING RECORD BOOK FOR RATINGS FORMING PART OF AN ENGINEERING WATCH AND RATINGS QUALIFYING AS ABLE SEAFARER ENGINE

INTRODUCTION

In 2010, the IMO Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW) was revised and updated. This ISF Training Record Book takes full account of these new requirements for ratings forming part of an engineering watch. This Book also addresses the additional competence standards stipulated by IMO for the grade of Able Seafarer Engine.

The revised STCW Convention continues to place emphasis on assessment of the outcome of training, i.e. the ability of seafarers to perform their duties competently. In particular, the 2010 Convention requires that a rating's seagoing service must be properly structured and recorded in a training record book approved by the maritime administration responsible for issuing certificates of competence. The footnotes of the amended STCW text specifically refer to ISF training record books as an example of such documentation.

The STCW Code, which contains the detailed requirements of the revised STCW Convention, sets out new uniform standards for the attainment of competences in the various maritime skills required to qualify as a watchkeeping rating or an Able Seafarer Engine. The STCW Code also stipulates criteria by which a trainee's attainment of these competences should be assessed by designated on board training officers.

The tasks contained in this Record Book have been carefully designed to help ensure that trainees meet the requirements for certification stipulated by the "competences"[1] and that as far as possible the officers supervising their training use evaluation based on Table A-III/4 and Table A-III/5 of the STCW Code. However, the tasks have been arranged with on board training in mind. The training tasks and associated criteria are, in many instances, presented in more detail than in the text of the Convention. This is to help ensure that trainees make the best use of their seagoing service and to help officers supervising trainees make a more objective evaluation of whether they are indeed competent.

Completion of the ISF Training Record Book should provide sufficient documentary evidence that a trainee has completed a properly structured on board training programme and has demonstrated competence in the skills required by the amended STCW Convention in order to be certificated as a rating forming part of an engineering watch and/or as an Able Seafarer Engine.

Note: The certificate issuing administration, which may or may not be the same as the flag of the ship(s) on which on board training is conducted, may determine appropriate training, assessment and certification requirements in addition to the competences and tasks contained in the Record Book. In practice, however, most if not all of such requirements will form part of shore based Basic Training (in accordance with Section A-VI/1 of the STCW Code) and Special Training, as applicable (in accordance with Regulation III/4).

1 Additional guidance on the competence-based approach to training is given in the ISF Guidelines to the IMO STCW Convention, including the 2010 Manilla Amendments.

CONTENTS

SECTION 1 GUIDE TO COMPLETION

FOR THE ATTENTION OF CHIEF ENGINEERS, DESIGNATED ON BOARD TRAINING OFFICERS AND RATING TRAINEES

Object of the Record Book

The purpose of this Record Book is to help ensure that ratings follow a structured programme of training and make best use of their time at sea. In so doing they will gain the practical training and experience necessary to become competent watchkeeping engine ratings, and/or Able Seafarers, in accordance with the STCW Convention as amended in 2010. It is therefore important that this guidance is carefully followed.

On receipt of this Book:

The trainee should complete the information required on the following pages, including details of Basic Training received in accordance with the STCW Convention (page 7). The trainee will then be personally responsible for the safe keeping of this Record Book throughout training.

- Section 3, concerning details of mandatory Safety Familiarisation (page 16) and mandatory Shipboard Familiarisation (page 17), should be completed **immediately** after the trainee joins each ship. An officer should sign to signify that mandatory familiarisation as required by the STCW Convention has been undertaken.

As soon as possible after joining each ship:

- The trainee should complete Section 4 (page 19) concerning the technical details of the vessel. The master and the designated training officer on board each ship should provide an opportunity for this exercise to be undertaken.

- The designated on board training officer appointed by the chief engineer should inspect this Book in order to check progress already made. A plan should be made to tackle the competences that still need to be demonstrated. A Task Summary Chart can be found in Section 8 (page 68).

Throughout the rating's seagoing service:

- Section 6 and 7, which contain a list of on board training tasks, should be progressively completed. Additional guidance on recording progress is given at the start of Section 5 (page 22).

- The Task Summary Chart in Section 8 (page 68) should also be progressively completed.

- The Book should be submitted to the chief engineer for inspection every month and at the end of each voyage. The chief engineer's comments should be recorded, dated and stamped (page 13). Comments should only relate to the trainee's competence and practical progress.

- The Book should be submitted to the designated on board training officer on joining each vessel – and then, so far as the voyage pattern allows, every week. Comments should be recorded (page 10).

- A precise record should be kept of the trainee's seagoing service including time spent on engine watchkeeping duties (page 9).

Important note

The STCW Convention requires that any person conducting on board training shall do so only when it will not adversely affect the normal operation of the ship and time can be dedicated to the training and any evaluation of competence.

SECTION 2 SUMMARY RECORD OF PROGRESS

PARTICULARS OF TRAINEE to be completed by the trainee in **BLOCK CAPITALS**

Name in Full ....

Seafarer's Book No. .... Date of Birth ....

Home Address ....

....

....

Change of Address (if applicable) ....

....

....

PHOTO

Sponsoring Company ....

Address ....

....

....

SPECIAL TRAINING as applicable in accordance with Regulation III/4 of the STCW Convention

This Special Training, if undertaken, may result in a reduction of the minimum required seagoing service that must precede the issue of an STCW certificate for watchkeeping at the support level. Such training is in addition to Basic Training (see below) and as a minimum should normally cover the underpinning knowledge required to perform competences contained in Section A-III/4 of the STCW Code (see Section 5 of this Book). If uncertain, your company should be able to advise you of the seagoing service required.

COLLEGE PHASES:		
	From:	To:
	From:	To:
	From:	To:
	From:	To:
SEA PHASES:		
	From:	To:
	From:	To:
	From:	To:
	From:	To:

BASIC TRAINING as required by Section A-VI/1 of the STCW Code

As part of your pre-sea training you should have completed Basic Training or instruction as listed. Enter details of this training or instruction below.

	Date	Location	Document Number
Personal Survival Techniques			
Fire Prevention and Fire Fighting			
Elementary First Aid			
Personal Safety and Social Responsibilities			

RECORD OF OTHER TRAINING

	Date	Location	Document Number
Proficiency in Survival Craft & Rescue Boats			
Proficiency in Fast Rescue Boats			
Security Training			

SHIPBOARD RECORD OF SERVICE

Ship	IMO Number	Dates		Time Spent on Engine Watchkeeping Functions		Voyage Total - Seagoing Service	
		Sign on	Sign off	Months	Days	Months	Days
			Total Service				

DESIGNATED TRAINING OFFICER'S REVIEW OF TRAINING PROGRESS

This table should be completed weekly or at such intervals as the trading of the vessel allows.
Comments should only relate to the trainee's practical progress and competence and should NOT refer to character.

Ship	Comments	Name in BLOCK CAPITALS	Initials	Date

DESIGNATED TRAINING OFFICER'S REVIEW OF TRAINING PROGRESS (CONTINUED)

Ship	Comments	Name in BLOCK CAPITALS	Initials	Date

DESIGNATED TRAINING OFFICER'S REVIEW OF TRAINING PROGRESS (CONTINUED)

Ship	Comments	Name in BLOCK CAPITALS	Initials	Date

CHIEF ENGINEER'S INSPECTION OF RECORD BOOK

Comments should only relate to the trainee's practical progress and competence and should NOT refer to character.

Ship	Comments	Name in BLOCK CAPITALS	Initials	Date	Ship's Official Stamp

COMPANY'S INSPECTION OF RECORD BOOK

Comments should only relate to the trainee's practical progress and competence and should NOT refer to character.

Comments	Name in BLOCK CAPITALS	Initials	Date

LIST OF PUBLICATIONS, VIDEO OR COMPUTER-BASED TRAINING PROGRAMMES STUDIED/USED

Date	Subject/Title	Officer's Initials

SECTION 3 MANDATORY SAFETY AND SHIPBOARD FAMILIARISATION

SAFETY FAMILIARISATION as required by Section A-VI/1 paragraph 1 of the STCW Code

Before being assigned to shipboard duties all seafarers must receive basic safety familiarisation to know what to do in an emergency. The chief engineer or responsible officer on each ship should sign and date below to signify that you have received training or instruction to be able to carry out the following tasks or duties.

Ship's Name			
Task/Duty	**Officer's Initials/Date**	**Officer's Initials/Date**	**Officer's Initials/Date**
Be able to: Communicate with other persons on board on elementary safety matters			
Understand safety information symbols, signs and alarm signals			
Know what to do if: A person falls overboard Fire or smoke is detected The fire or abandon ship alarm is sounded			
Be able to: Identify muster and embarkation stations and emergency escape routes			
Locate and don life jackets and survival suits			
Raise the alarm and have a basic knowledge of the use of portable fire extinguishers			
Take immediate action upon encountering an accident or other medical emergency before seeking further medical assistance on board			
Close and open the fire, weathertight and watertight doors fitted in the particular ship, other than those for hull openings			

SHIPBOARD FAMILIARISATION as required by Regulation I/14 of the STCW Convention

You will be given a period of time during which you will have an opportunity to become acquainted with the equipment you will be using, and specific watchkeeping, safety, environmental and emergency procedures and arrangements required to perform your duties. The location of safety and emergency equipment varies from ship to ship. To be sure that you are familiar with your duties and all ship arrangements, installations, equipment procedures and ship characteristics that are relevant to your routine or emergency duties, you must complete the following tasks or duties as soon as possible on joining your ship.

Ship's Name			
Task/Duty	**Officer's Initials/Date**	**Officer's Initials/Date**	**Officer's Initials/Date**
Watckeeping procedures and arrangements: Visit bridge, look-out post, forecastle, poopdeck, main deck and other work areas			
Get acquainted with steering controls, telephones, telegraphs and other bridge equipment and displays			
Activate, under supervision, equipment to be used in routine duties			
Safety and emergency procedures: Read and demonstrate an understanding of your Company's Fire and Safety Regulations			
Demonstrate recognition of the alarm signals for: **FIRE** **EMERGENCY** **ABANDON SHIP**			
Locate medical and first aid equipment			
Locate fire fighting equipment: alarm activating points, alarm bells, extinguishers, hydrants, fire axes and hoses			
Locate rocket line throwing apparatus			
Locate distress rockets, flares and other pyrotechnics			
Locate breathing apparatus and firefighter's outfits etc			
Locate EPIRB, SART and walkie-talkie radios			
Locate Emergency Escape Breathing Devices (EEBDs)			
Locate CO_2 bottle room, and control valves for smothering apparatus in machinery spaces, pump rooms, cargo tanks and holds			
Locate and understand the operation of the emergency pump			
Environmental protection: Get acquainted with: The procedure for handling garbage, rubbish and other wastes			
The use of garbage compactor or other equipment as appropriate			

SHIPBOARD FAMILIARISATION as required by Regulation I/14 of the STCW Convention (continued)

Insert Boat and Fire Muster Stations and other details in the appropriate space. Ask the master to sign in the space provided.

Ship's Name			
Boat Muster Station			
Fire Muster Station			
Master's Name in BLOCK CAPITALS			
Master's Signature			
Date			

SECTION 4 PARTICULARS OF SHIPS

It is an essential feature of your training that you obtain knowledge of the ships on which you serve. To assist you in meeting this important requirement the following particulars are to be recorded during the time spent on each ship.

FIRST SHIP

mv/ss	**IMO Number**...	**Call Sign** ....
Dimensions and Capacities	**Generators**	**Cargo Handling Gear**
Length Over All .... m	Engines (no. / type)	Derricks/cranes (no. and SWL) .... tonnes
Breadth .... m	Max electrical output .... kW	Winches (types) .... tonnes
Depth .... m	Emergency generator engine (type)	Other cargo equipment..
Summer draft .... m	Max electrical output	....
Summer freeboard .... m	**Lifesaving equipment**	....
Gross tonnage .... t	Fast Rescue Boat (engine / type)	....
Deadweight .... t	Lifeboats (no. / cap / engine)	Ballast tanks (no.)...
Light displacement... t	Life-rafts (no. / capacity)	Cargo tanks (no.)..
Grain/Liquid capacity .... m^3	Lifebuoys (no.)	Cargo pumps (no.)..
Main Engines	Survival suits (no. / type)	Pipelines (sizes)..
Engine (make / type)	Escape sets (no. / type)	Type and rating .... tonnes/hour
MCR .... kW at .... revs per min	**Fire Fighting Equipment**	**Steering and Propulsion**
Boilers (make / type)	Emergency fire pump (type / cap)	Steering gear (make / type) ....
HFO capacity .... m^3/tonnes	Fire extinguishers (number and capacity)	Propeller (type) ....
DO capacity .... m^3/tonnes	Types: Water .... litres	Propeller (no. of blades / diameter) ....
HFO consumption .... tonnes/day	Foam .... litres	Thrusters (no. / type / thrust) ....
DO consumption .... tonnes/day	Dry powder .... kg	**Anchors**
Service speed .... knots	CO_2 .... kg	Port .... tonnes
Fresh water generators (tonnes / day)	Fire hoses (no. and size) .... mm	Starboard .... tonnes
	Breathing apparatus (no. / make)	

SECOND SHIP

mv/ss ....

IMO Number ....

Call Sign ....

Dimensions and Capacities

Length Over All .... m

Breadth .... m

Depth .... m

Summer draft .... m

Summer freeboard .... m

Gross tonnage .... t

Deadweight .... t

Light displacement .... t

Grain/Liquid capacity .... m^3

Main Engines

Engine (make / type) ....

MCR .... kW at .... revs per min

Boilers (make / type) ....

HFO capacity .... m^3/tonnes

DO capacity .... m^3/tonnes

HFO consumption .... tonnes/day

DO consumption .... tonnes/day

Service speed .... knots

Fresh water generators (tonnes / day) ....

Generators

Engines (no. / type) ....

Max electrical output .... kW

Emergency generator engine (type) ....

Max electrical output ....

Lifesaving equipment

Fast Rescue Boat (engine / type) ....

Lifeboats (no. / cap / engine) ....

Life-rafts (no. / capacity) ....

Lifebuoys (no.) ....

Survival suits (no. / type) ....

Escape sets (no. / type) ....

Fire Fighting Equipment

Emergency fire pump (type / cap) ....

Fire extinguishers (number and capacity)

Types: Water .... litres

Foam .... litres

Dry powder .... kg

CO_2 .... kg

Fire hoses (no. and size) .... mm

Breathing apparatus (no. / make) ....

Cargo Handling Gear

Derricks/cranes (no. and SWL) .... tonnes

Winches (types) .... tonnes

Other cargo equipment ....

....

....

....

Ballast tanks (no.) ....

Cargo tanks (no.) ....

Cargo pumps (no.) ....

Pipelines (sizes) ....

Type and rating .... tonnes/hour

Steering and Propulsion

Steering gear (make / type) ....

Propeller (type) ....

Propeller (no. of blades / diameter) ....

Thrusters (no. / type / thrust) ....

Anchors

Port .... tonnes

Starboard .... tonnes

THIRD SHIP

mv/ss	**IMO Number**	**Call Sign** ....
Dimensions and Capacities	**Generators**	**Cargo Handling Gear**
Length Over All .... m	Engines (no./type)	Derricks/cranes (no. and SWL) .... tonnes
Breadth .... m	Max electrical output ....kW	Winches (types) .... tonnes
Depth .... m	Emergency generator engine (type)	Other cargo equipment ....
Summer draft .... m	Max electrical output	....
Summer freeboard .... m	**Lifesaving equipment**	....
Gross tonnage .... t	Fast Rescue Boat (engine/type)	....
Deadweight .... t	Lifeboats (no./cap/engine)	Ballast tanks (no.) ....
Light displacement .... t	Life-rafts (no./capacity)	Cargo tanks (no.) ....
Grain/Liquid capacity .... m^3	Lifebuoys (no.)	Cargo pumps (no.) ....
Main Engines	Survival suits (no./type)	Pipelines (sizes) ....
Engine (make/type)	Escape sets (no./type)	Type and rating .... tonnes/hour
MCR .... kW at .... revs per min	**Fire Fighting Equipment**	**Steering and Propulsion**
Boilers (make/type)	Emergency fire pump (type/cap)	Steering gear (make/type) ....
HFO capacity .... m^3/tonnes	Fire extinguishers (number and capacity)	Propeller (type) ....
DO capacity .... m^3/tonnes	Types: Water .... litres	Propeller (no. of blades/diameter) ....
HFO consumption .... tonnes/day	Foam .... litres	Thrusters (no./type/thrust) ....
DO consumption .... tonnes/day	Dry powder .... kg	**Anchors**
Service speed .... knots	CO_2 .... kg	Port .... tonnes
Fresh water generators (tonnes/day)	Fire hoses (no. and size) .... mm	Starboard .... tonnes
	Breathing apparatus (no./make)	

SECTION 5 INFORMATION ON TRAINING TASKS AND COMPETENCES TO BE ACHIEVED

Sections 6 and 7 of your Record Book give details of the training tasks you should complete as part of your on board training. **Section 7 should only be completed by trainees seeking to qualify as an Able Seafarer Engine.**

You will see that each page lists the tasks or duties you should undertake. Completion of these will lead to meeting the competences specified by the IMO STCW Convention. A senior officer should review your progress and indicate, with initials and date in the **green** box on the right hand side of the page, that your performance is considered to meet the *Criteria for Evaluation* and that competence has been demonstrated in that element. The officer may offer advice on areas in which improvement is necessary. The competences required by a watchkeeping rating as tabulated in the STCW Code are listed below.

The Sections are organised as follows:

SECTION 6 COMPETENCES FOR RATINGS FORMING PART OF A WATCH IN A MANNED ENGINE-ROOM OR DESIGNATED TO PERFORM DUTIES IN A PERIODICALLY UNMANNED ENGINE-ROOM (STCW CODE TABLE A-III/4):

SECTION 7 COMPETENCES FOR RATINGS AS ABLE SEAFARER ENGINE (STCW CODE TABLE A-III/5):

EXAMPLE OF HOW TO COMPLETE THE LIST OF TRAINING TASKS AND COMPETENCES ACHIEVED

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
1.	**Competence: Carry out a watch routine appropriate to the duties of a rating forming part of an engine-room watch** **Understand orders and be understood in matters relevant to watchkeeping duties**					
1.1	**Use correct terms in machinery spaces and names of machinery and equipment**			*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood*	CM	14/9/11
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Record information, as instructed, in the engine-room log book, engine movements log and other record books	AB	1/05/11	Write clearly in the columns and spaces provided. Take more care with readings and calculations. Try to improve your writing!	AB	2/06/11
.2	Observe and note normal operating temperatures/pressures	CM	12/6/11		CM	19/6/11
.3	Demonstrate correct use of terms used in the engine-room and names of machinery, equipment and tools	HC	17/6/11	Try to remember normal conditions so that you can immediately spot a fall off in performance.		

- The competence shaded in **green** (in this case 'Carry out watch routine etc') is directly taken from the text of the STCW Code.
- The primary task (in this case 'Use correct terms in machinery spaces etc') is also taken from the STCW Code. The primary task is sub divided into training tasks or duties on the left hand side of the page. The trainee should complete as many of these training tasks as possible. However, in view of the likely equipment, cargo or voyage pattern of the ship, **it is not expected that all these tasks will be completed before the trainee is considered competent in the primary task**. It should be noted that some of the skills and knowledge that underpin the competences may well have been obtained during shore based training.
- Space is provided to record completion of each training task twice by the supervising officer. This does not mean that each task must be completed twice if in the opinion of the officer, once is considered sufficient.
- The officer supervising the trainee does not necessarily have to be the designated training officer.
- Before 'Competence Demonstrated' is recorded (in the light green box), the chief engineer or designated on board training officer may record any appropriate advice about areas for improvement. A large blank space for this purpose is provided beneath the criteria for evaluation. As competence in these primary tasks is demonstrated, the appropriate light green boxes next to the criteria for evaluation, on the far right hand side of the page, should be signed and dated (day, month and year) by the chief engineer or designated training officer on board the ship to attest that competence has been demonstrated.
- A trainee's attainment of the competences should only be recorded as 'Competence Demonstrated' when the chief engineer or designated training officer is satisfied that the trainee can perform the duty without supervision.
- When recording 'Competence Demonstrated', careful account should be taken of the criteria for evaluation contained on the right hand side of the page, which is taken from the STCW Code, as well as the ordinary practices of seafarers and good safe working practices.

SECTION 6 TASKS FOR SUPPORT LEVEL RATINGS FORMING PART OF AN ENGINEERING WATCH

This section can be completed on its own for certification as an engine rating forming part of an engineering watch, or can be completed in combination with the training set out in Section 7 for certification as an Able Seafarer Engine.

The training in this section of this record book covers the requirements for the certification of ratings forming part of an engineering watch. The competences covered by Section A-III/4 of the STCW Code are limited to the function Marine Engineering at the Support Level.

The requirements for certification are set out in Regulation III/4:

Regulation III/4
Mandatory minimum requirements for certification of ratings forming part of a watch in a manned engine-room or designated to perform duties in a periodically unmanned engine-room

1 Every rating forming part of an engine-room watch or designated to perform duties in a periodically unmanned engine-room on a seagoing ship powered by main propulsion machinery of 750kW propulsion power or more, other than ratings under training and ratings whose duties are of an unskilled nature, shall be duly certificated to perform such duties.

2 Every candidate for certification shall:

- **.1** be not less than 16 years of age;
- **.2** have completed:
  - **.2.1** approved seagoing service including not less than six months of training and experience, or
  - **.2.2** special training, either pre-sea or on board ship, including an approved period of seagoing service which shall not be less than two months; and
- **.3** meet the standard of competence specified in section A-III/4 of the STCW Code.

3 The seagoing service, training and experience requirement by subparagraphs 2.2.1 and 2.2.2 shall be associated with engine-room watchkeeping functions and involve the performance of duties carried out under the direct supervision of a qualified engineer officer or a qualified rating.

Completion of the ISF Training Record Book for engine ratings should ensure a structured approach is undertaken so that trainees can first qualify as ratings forming part of an engineering watch. However, completing Sections 6 and 7 together might assist in the early certification as an Able Seafarer Engine.

A Task Summary Chart, which trainees can tick off as they complete their on board training tasks, can be found in Section 8.

FUNCTION: MARINE ENGINEERING AT THE SUPPORT LEVEL

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
1.	**Competence: Carry out a watch routine appropriate to the duties of a rating forming part of an engine-room watch** **Understand orders and be understood in matters relevant to watchkeeping duties**					
1.1	**Use correct terms in machinery spaces and names of machinery and equipment**			*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Record information, as instructed, in the engine-room log book, engine movements log and other record books					
.2	Observe and note normal operating temperatures/pressures					
.3	Demonstrate correct use of terms used in the engine-room and names of machinery, equipment and tools					
1.2	**Conduct engine-room watchkeeping procedures**			*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood. Maintenance, handover and relief of the watch is in conformity with accepted principles and procedures*		
.1	Assist with preparing machinery for a sea passage					
.2	Assist with checking the starting air compressor and prepare starting air systems					
.3	Assist with the duties of an engineer officer on: Seagoing watches					
.4	Port/anchor watches					
.5	Accept and handover the watch duties in a timely manner					
.6	Under supervision carry out all routine watchkeeping duties, checking the correct functioning of automatic control and monitoring systems					

.7	Make adjustments as found necessary					
.8	Under supervision clean main engine cooling water suctions					
.9	Understand the need for high and low level suctions and changeover procedures					
.10	Check efficiency of sheathing on high-pressure fuel pipes and associated leak-off indicators					
.11	Perform routine checks and top ups to maintain cooling water system tanks and boiler hot well at the correct levels					
.12	Perform routine checks and top ups to maintain fuel oil and lube oil system tanks at the correct levels					
.13	Under supervision blow down scavenge drains					
.14	Ensure that automatic drains are functioning correctly					
.15	Assist with boiler water tests and treatment					
.16	Check the correct operation of the boiler including water level and burner					
.17	Assist with soot-blowing or associated procedure					
.18	Assist with preparing an evaporator/fresh water generator					
.19	Check all air receiver drains					
.20	Check correct operation of: Fuel oil purifiers and filters					
.21	Lube oil purifiers and filters					
.22	Assist with change over of lube oil coolers					
.23	Assist with change over of fresh water coolers					
.24	Assist with a tank cleaning operation					
.25	Assist watch officer on standby departure					
.26	Assist watch officer on standby arrival					

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
1.	**Competence: Carry out a watch routine appropriate to the duties of a rating forming part of an engine-room watch** **Understand orders and be understood in matters relevant to watchkeeping duties**				
1.2	**Conduct engine-room watchkeeping procedures (continued)**		*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood. Maintenance, handover and relief of the watch is in conformity with accepted principles and procedures*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.27	Assist with the change over to standby duties for: Main engines				
.28	Generators				
.29	Pumps				
.30	Steering gear				
.31	Assist with preparations for manoeuvring				
.32	Assist with changeover from heavy fuel oil to distillate operation				
.33	Assist with changeover from distillate to heavy fuel oil operation				
.34	Assist with shutting down main engine and auxillary systems when finished with engines				
.35	Understand and execute orders in English concerning: Routine operations				
.36	Emergency situations				
.37	Assist with make up of brine, if appropriate				
.38	Check density of brine				
.39	Assist with replacing desiccant and filters				

.40	Assist with putting sewage system on line					
.41	Assist with setting up: Ballast water system for pumping					
.42	Bilge pump					
.43	Assist with: Receiving bunkers					
.44	Transfer of fuel from bunker tanks to service tanks					
.45	Assist an officer with: Deballasting/ ballasting/ stripping tanks					
.46	Opening units, cleaning parts and reassembly					
1.3	**Use safe working practices as related to engine-room operations**			*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood*		
.1	Assist with preparation and testing the steering gear and telegraphs for sea					
.2	Understand the purpose of the crankcase oil mist detector					
.3	Identify any special precautions to be taken for electrical maintenance in hazardous areas					
.4	Demonstrate an understanding of safe working practices and procedures for use of power operated tools					
.5	Demonstrate an understanding of safe working practices and procedures for entry into enclosed spaces (tanks, duct keels, cable and pipe trunks, fridges, cargo holds, access trunks and other hazardous spaces)					
.6	Demonstrate an understanding of safe working practices and procedures for work beneath floor plates					
.7	Demonstrate an understanding of safe working practices and procedures for use of lifting gear and correct rigging, slinging and eyebolts					
.8	Demonstrate an understanding of safe working procedures involved in moving heavy machinery					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
1.	**Competence: Carry out a watch routine appropriate to the duties of a rating forming part of an engine-room watch** **Understand orders and be understood in matters relevant to watchkeeping duties**					
1.3	**Use safe working practices as related to engine-room operations (continued)**			*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.9	Demonstrate an understanding of safe working practices and procedures: Work within refrigeration machinery spaces					
.10	Inert Gas machinery spaces					
.11	Work on electrical machinery, including signage, lock out and tagging					
.12	Disposal of oily waste materials					
.13	Use of appropriate protective clothing					
.14	Hot work and hot work permit					
.15	Marking off work areas with blocking tape or fencing					
.16	Demonstrate an understanding of the precautions required for working on staging					
.17	Demonstrate an understanding of warning, mandatory safety and prohibition signs					
.18	Demonstrate an understanding of commonly used markings identifying piping systems in accordance with content or function					
.19	Assist with testing of electrical emergency stops					
.20	Assist with the opening, closing and securing of engine-room and machinery hatches					

.21	Assist with the testing of watertight doors, ports, hatches and quick closing valves					
.22	Ensure that all gear, tools, spares etc are properly stowed and secured					
.23	Assist with the rigging of safety lines and guard rails					
1.4	**Practice basic environmental protection procedures**			*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood*		
.1	Assist with setup and use of oily water separator					
.2	Assist with bilge pumping in compliance with MARPOL					
.3	Operate waste shredder/compactor/incinerator					
.4	Dispose of garbage ashore/at sea in compliance with MARPOL					
.5	Assist with disposal of sludge to shore reception facilities in compliance with MARPOL					
.6	For above duties, be aware of record keeping requirements for all operations					
1.5	**Use of appropriate internal communication system**			*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood*		
.1	Demonstrate the reporting procedures when entering, working in and leaving an otherwise unmanned machinery space (UMS)					
.2	Demonstrate correct use of internal communication systems					

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
1.	**Competence: Carry out a watch routine appropriate to the duties of a rating forming part of an engine-room watch** **Understand orders and be understood in matters relevant to watchkeeping duties**				
1.6	**Test engine-room alarm systems and demonstrate ability to distinguish between the various alarms, with special reference to fire extinguishing gas alarms**		*Communications are clear and concise and advice or clarification is sought from the officer of the watch where watch information or instructions are not clearly understood*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Assist with the testing of the engine-room fire detection system, and demonstrate an understanding of the purpose for which the different types of detector are fitted				
.2	Assist with the testing of the engine-room fixed fire extinguishing system gas release valve				
.3	Be aware of the safety precautions required prior to testing either of the above systems				
.4	Be aware of the safety precautions required prior to release of a fire extinguishing gas into the engine-room				
.5	Test and be able to identify the different alarms and understand actions to be taken when these alarms operate: General alarm Fire alarm High level alarm Low level alarm OWS alarm Gas release alarm Water tight door closing alarm Other engine-room alarms (list)				
.6	Where fixed CO_2 system is fitted for fire extinguishing, demonstrate knowledge of procedures for use and recognise alarms				
.7	Where water drench system is fitted, demonstrate knowledge of procedures for use and recognise alarms				

On ships where a boiler watch is maintained ratings should demonstrate a knowledge of procedures for the safe operation of boilers and associated equipment and demonstrate ability to maintain correct water levels and steam pressures; and the necessity for cleanliness and good housekeeping practice in the boiler area.

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
2.	**Competence: Maintain the correct levels and steam pressures (for keeping a boiler watch)**					
2.1	**Safe operation of boilers**			*Assessment of boiler condition is accurate and based on relevant information available from local and remote indicators and physical inspections. The sequence and timing of adjustments maintains safety and optimum efficiency*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Assist with raising the temperature of fuel oil from cold to the correct level					
.2	Assist with filling a boiler and raising steam from cold					
.3	Assist with changeover from heavy fuel oil to distillate operation					
.4	Assist with changeover from distillate to heavy fuel oil operation					
.5	Admit steam to a line or system, taking all precautions against thermal and pressure shock and the avoidance of water hammer					
.6	Check the security of steam pipes and provision for expansion					
.7	Check that steam traps and drains are functioning					
.8	Close down a steam line, observing procedure for draining					
.9	Check quality of combustion, noting: Smoke from the funnel Clarity around the flame Flame shape, size and colour Excess air, CO_2/CO readings Carbon and unburnt fuel deposits					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
2.	**Competence: Maintain the correct levels and steam pressures (for keeping a boiler watch)**					
2.1	**Safe operation of boilers (continued)**			*Assessment of boiler condition is accurate and based on relevant information available from local and remote indicators and physical inspections. The sequence and timing of adjustments maintains safety and optimum efficiency*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.10	Check returns from heating coils and other possible sources of contaminated feedwater					
.11	Assist with checking the correct functioning of all boiler condition indicators and alarms					
.12	Assist with changing and overhauling burners					
.13	Check that correct boiler water level is maintained					
.14	Understand the effect of varying the temperature of circulating water					
.15	Assist with taking a boiler out of service					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
3.	**Competence: Operate emergency equipment and apply emergency procedures**					
3.1	**Demonstrate emergency duties**			*Initial action on becoming aware of an emergency or abnormal situation conforms with established procedures. Communications are clear and concise at all times and orders are acknowledged in a seamanlike manner*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Take correct action during emergency drills: Fire drill					
.2	Black out drill					
.3	Abandon ship drill					
.4	Rescue from an enclosed space					
.5	Assist with use of main engine local control and emergency manoeuvring					
.6	Assist with procedure for returning main engine to normal running					
.7	Assist with a drill for emergency running procedures					
.8	Demonstrate knowledge of steering gear emergency operation					
.9	Participate in an emergency response exercise for controlling spillage of oil on board					
.10	Participate in drill for clean-up of hazardous cargo spillage					
.11	Participate in an emergency response exercise for water ingress, e.g. in case of collision, stranding or grounding					
.12	Explain the operation of rocket line throwing apparatus					
.13	Assist with the inspection and maintenance of life saving appliances, launching davits and gear					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
3.	**Competence: Operate emergency equipment and apply emergency procedures**					
3.1	**Demonstrate emergency duties (continued)**			*Initial action on becoming aware of an emergency or abnormal situation conforms with established procedures. Communications are clear and concise at all times and orders are acknowledged in a seamanlike manner*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.14	Assist with the routine maintenance of a lifeboat engine					
.15	Participate in an emergency first aid drill at sea					
.16	Demonstrate a basic understanding of first aid principles: Stopping bleeding					
.17	Treatment of suffocation/drowning					
.18	Placing casualty in the recovery position					
.19	Demonstrate how to handle a casualty in shock					
.20	Demonstrate procedures for dealing with heat stroke					
.21	State procedure for dealing with a casualty of electric shock					
.22	Demonstrate procedure for treating burns					
.23	Demonstrate use of safety harness and line					
3.2	**Use escape routes from machinery spaces**			*Initial action on becoming aware of an emergency or abnormal situation conforms with established procedures. Communications are clear and concise at all times and orders are acknowledged in a seamanlike manner*		
.1	Locate and demonstrate use of the escape routes from machinery spaces					
.2	Identify location of all emergency escape breathing devices (EEBDs)					

.3	Demonstrate donning and use of emergency escape breathing device set					
.4	Participate in a search and rescue drill from a machinery space					
3.3	**Demonstrate familiarity with the location and use of fire-fighting equipment in the machinery spaces**			*Initial action on becoming aware of an emergency or abnormal situation conforms with established procedures. Communications are clear and concise at all times and orders are acknowledged in a seamanlike manner*		
.1	Assist with routine testing and maintenance of: Fire pumps					
.2	Fire flaps					
.3	ER fire extinguishing system and equipment					
.4	Understand use and assist in the maintenance of: Portable foam extinguisher					
.5	Portable CO_2 extinguisher					
.6	Portable dry powder extinguisher					
.7	Perform fire patrol duties					
,8	Participate in an emergency response exercise for fire at sea and in port					
.9	Assist with the testing of the following systems, where fitted: Fixed automatic sprinklers					
.10	Fixed steam systems					
.11	Fixed foam extinguishers					
.12	Fixed CO_2 systems					
.13	Fire flaps and dampers					
.14	Automatic and manual fire doors					
.15	Emergency shut off valves, pump stops and main engine stops					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
3.	**Competence: Operate emergency equipment and apply emergency procedures**					
3.3	**Demonstrate familiarity with the location and use of fire-fighting equipment in the machinery spaces (continued)**			*Initial action on becoming aware of an emergency or abnormal situation conforms with established procedures. Communications are clear and concise at all times and orders are acknowledged in a seamanlike manner*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.16	Demonstrate use of equipment in the fire-fighter's outfit					
.17	Describe the operation of the fixed fire extinguishing system					
.18	Demonstrate the procedures and precautions required for entry into an enclosed space					
.19	Demonstrate donning and use of breathing apparatus sets					
.20	Demonstrate donning and use of protective clothing and equipment					

SECTION 7 TASKS FOR RATINGS QUALIFYING AS ABLE SEAFARER ENGINE

The training tasks for Able Seafarer Engine which follow may be completed alongside watch rating tasks for Marine Engineering at the Support Level in Section 6. This may assist in the early achievement of Able Seafarer Engine certification.

The STCW competence standards for Able Seafarer Engine reflect the broader work of engine ratings and include additional competences in engineering plus competences for three additional Support Level functions:

- Electrical, Electronic and Control Engineering
- Maintenance and Repair at the Support Level
- Controlling the Operation of the Ship and Care for Persons on Board

The requirements for certification of Able Seafarers Engine are set out in STCW Regulation III/5:

Regulation III/5
Mandatory minimum requirements for certification of ratings as Able Seafarer Engine in a manned engine-room or designated to perform duties in a periodically unmanned engine-room

1 Every Able Seafarer Engine serving on a seagoing ship powered by main propulsion machinery of 750kW propulsion power or more shall be duly certificated.

2 Every candidate for certification shall:

.1 be not less than 18 years of age;

.2 meet the requirements for certification as a rating forming part of a watch in a manned engine-room or designated to perform duties in a periodically unmanned engine-room;

.3 while qualified to serve as a rating forming part of an engineering watch, have approved seagoing service in the engine department of:

.3.1 not less than 12 months, or

.3.2 not less than 6 months and have completed approved training; and

.4 meet the standard of competence specified in section A-III/5 of the STCW Code.

A Task Summary Chart, which trainees can tick off as they complete their on board training tasks, can be found in Section 8.

FUNCTION: MARINE ENGINEERING AT THE SUPPORT LEVEL (ABLE SEAFARER ENGINE)

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
4.	**Competence: Contribute to a safe engineering watch**					
4.1	**Ability to understand orders and to communicate with the officer of the watch in matters relevant to watchkeeping duties**			*Communications are clear and concise*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Understand orders for testing and preparing for sea: Steering system Telegraphs Auxiliary plant and equipment Main engine manoeuvring (list others):					
.2	Understand and communicate orders					
.3	Understand the need to promptly seek clarification and confirmation if an officer's order is unclear or contrary to safe practice					
.4	Play an effective role as a team member at all times					
4.2	**Procedures for the relief, maintenance and handover of a watch**			*Maintenance, handover and relief of the watch is in conformity with acceptable practices and Safety Management System (SMS) procedures*		
.1	Consistently arrive on watch appropriately dressed, fit for duty and on time					
.2	Receive and pass on information relevant to watch duties clearly during: UMS control operation					
.3	Bridge control operation					

.4	Seek clarification if unsure on any matter when taking over watch (at sea, at anchor, in port)					
.5	Understand procedures to be followed during: UMS control operation					
.6	Bridge control operation					
.7	Check normal operation of automatic control and monitoring systems					
.8	Perform routine ER rounds and report or log rounds as appropriate					
.9	Understand the need to refuse to hand over the watch if there is reason to believe that the relief is not capable of carrying out watchkeeping duties effectively					
4.3	**Information required to maintain a safe watch**			*Maintenance, handover and relief of the watch is in conformity with acceptable practices and procedures*		
.1	In port, maintain watch as per instructions					
.2	Check starting air compressor and starting air system					
.3	Check for safety hazards on plates, walkways and access points					
.4	Assist with the opening, closing and securing of engine room and machinery hatches					
.5	At anchor, maintain watch as per instructions					
.6	Call for assistance if alarms or unusual demands arise from operational circumstances					
.7	Assist with shut down of main engines when finished with engines					
.8	Assist with shut down of auxiliary systems when finished with engines					
.9	Conduct boiler water tests and report results					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
5.	**Competence: Contribute to the monitoring and controlling of an engine-room watch**					
5.1	**Basic knowledge of the function and operation of main propulsion and auxiliary machinery**			*The frequency and extent of monitoring of main propulsion and auxiliary machinery conforms with accepted principles and procedures*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Assist with steering gear checks with bridge prior to operation					
.2	Trace out and describe the following systems: Main engine fuel system					
.3	Main engine lubricating oil system					
.4	Main sea water cooling system					
.5	Assist with preparations of engine room machinery for manoeuvring					
.6	Assist in manoeuvring the main engine from the engine-room control position, including starting, stopping and reversing					
.7	Understand the procedure for controlling the main engine from the emergency manoeuvring position(s), including start/stop/reverse or CPP or reverse clutch operation					
.8	Under supervision, respond to instructions from the bridge and operate the main engine controls during periods of manoeuvring					
.9	Carry out post start-up checks of main engine and shafting					
.10	Under supervision, adjust main engine and auxiliary machinery for sea passage					
.11	Under supervision, change over from HFO to distillate operations					
.12	Under supervision, change over from distillate to HFO operations					

.13	Trace out and describe the following systems: Auxiliary steam system					
.14	Diesel engine water cooling systems					
.15	Change over main and standby compressor to normal running mode					
.16	Change over to standby duties for pumps					
.17	Under supervision, start up sewage system					
.18	Conduct routine operations on potable fresh water service					
.19	Understand routine operations for pumping bilges					
.20	Check exhaust pipes for leakage: Main engines					
.21	Auxiliary engines					
.22	Emergency engines					
.23	Operate auxiliary heating boilers including combustion system					
.24	Assist with routine tests on: Engine cooling water					
.25	Fuel oil					
.26	Lube oil					
.27	Monitor operations of lube oil and fuel oil purifiers					
.28	Under supervision, conduct feedwater treatment					
.29	Adjust feedwater treatment according to test findings and instruction from a senior officer					
.30	Check the correct functioning of all boiler condition indicators and alarms					
.31	Engage, disengage and operate turning gear					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
5.	**Competence: Contribute to the monitoring and controlling of an engine-room watch**					
5.2	**Basic understanding of main propulsion and auxiliary machinery control pressures, temperature and levels**			*Deviations from the norm are identified. Unsafe conditions or potential hazards are promptly recognized, reported and rectified before work continues*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Carry out routine watchkeeping duties, checking the correct functioning of automatic control and monitoring systems					
.2	Monitor and report main engine cooling water temperature					
.3	Understand the range and deviation of temperatures of engine cooling water intake and outflow					
.4	Understand the need to maintain normal temperatures in circulating water					
.5	Assist with routine main engine jacket cooling water tests					
.6	Monitor and report main engine oil pressure and temperature					
.7	Understand the range and deviation of temperatures and pressures of engine lube oil systems					
.8	Have a knowledge of the system valve positions for normal running					
.9	Assist in carrying out routine testing of main engine safety trips and alarms					
.10	Check and report fuel service tank level and water content					
.11	Check and report auxiliary engine cooling water temperature					
.12	Check and report auxiliary engine oil pressure and temperature					

.13	Assist in carrying out routine testing of auxiliary boiler safety trips and alarms					
.14	Under supervision, blow down gauge glasses and perform checks on the correct boiler water level indication					
.15	Assist in preparations to run an evaporator/fresh water generator					
.16	Assist in tests and conditioning for purity of potable fresh water					
.17	Check normal operation of ship's AC plant					
.18	Assist with shut down and secure refrigeration/AC plant					
.19	Carry out correct cleaning procedures for AC filters					
.20	Understand the need for logging (in accordance with MARPOL) of any refrigerant gas recharge					
.21	Check crankcase oil mist detector and understand the need for action to be taken in case of an alarm					
.22	Demonstrate a knowledge of the normal operations of governors					
.23	Demonstrate a knowledge of pressure tank safety devices					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
6.	**Competence: Contribute to fuelling and oil transfer operations**					
6.1	**Knowledge of the function and operation of fuel system and oil transfer operations, including preparations for fuelling and transfer operations**			*Transfer operations are carried out in accordance with established safety practices and equipment operating instructions*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Open up lines to tanks in preparation for fuel transfer					
.2	Use cross connections, where appropriate					
.3	Conduct routine operations for lining up and receiving fuel oil					
.4	Have knowledge of and observe pollution prevention requirements					
.5	For an internal fuel transfer, prepare for and carry out fuel oil transfer, as instructed					
6.2	**Knowledge of the function and operation of fuel system and oil transfer operations, including procedures for connecting and disconnecting fuelling and transfer hoses**			*The handling of dangerous, hazardous and harmful liquids complies with established safety practices. Communications within the operator's area of responsibility are consistently successful*		
.1	As a team member, check correct connection and disconnection of fuel transfer hose					
.2	Check MARPOL sampler is correctly connected and sample taken					
.3	As a team member, drain transfer hose before disconnection					

6.3	**Knowledge of the function and operation of fuel system and oil transfer operations, including procedures relating to incidents that may arise during fuelling or transferring operation**			*The handling of dangerous, hazardous and harmful liquids complies with established safety practices. Communications within the operator's area of responsibility are consistently successful*		
.1	Demonstrate a knowledge of spill containment and clean up procedures during transfer operations					
.2	Raise the temperature of fuel oil in storage tanks from cold to the correct level					
.3	Demonstrate shut down					
.4	Use holding tanks					
6.4	**Knowledge of the function and operation of fuel system and oil transfer operations, including securing from fuelling and transfer operations**			*The handling of dangerous, hazardous and harmful liquids complies with established safety practices. Communications within the operator's area of responsibility are consistently successful*		
.1	Secure and tag valves on completion of bunkering					
.2	Secure blank on deck manifold					
6.5	**Knowledge of the function and operation of fuel system and oil transfer operations, including ability to correctly measure and report tank levels**			*The handling of dangerous, hazardous and harmful liquids complies with established safety practices. Communications within the operator's area of responsibility are consistently successful*		
.1	Sound bilges and record information					
.2	Take fuel oil tank ullages and record tank levels using gauging equipment					
.3	Using tape take fuel oil soundings and record tank levels					
.4	Transfer fuel oil from bunker tanks to service tanks to correct levels as instructed					
.5	Drain water/sludge from settling tanks					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
7.	**Competence: Contribute to bilge and ballast operations**					
7.1	**Knowledge of the safe function, operation and maintenance of the bilge and ballast systems, including reporting incidents associated with transfer operations**			*Operations and maintenance are carried out in accordance with established safety practices and equipment operating instructions and pollution of the marine environment is avoided. Communications within the operator's area of responsibility are consistently successful*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Recognise that clean ballast water tanks may contain oily water due to leakage or unknown contamination					
.2	Understand that any oil or oily water pumped or leaking into harbour/dock must be promptly reported					
.3	Understand that tank transfers should only be executed on the instructions of an officer					
.4	Assist in the operation of the bilge pumping system including the oily water separator, and assist in completion of necessary records					
.5	Check and clean tank suctions, strum boxes, bilges and wells					
7.2	**Knowledge of the safe function, operation and maintenance of the bilge and ballast systems, including ability to correctly measure and report tank levels**			*Communications within the operator's area of responsibility are consistently successful*		
.1	Sound bilges and record information					
.2	Take and report tank soundings manually and using gauging system					
.3	Assist in pumping bilges in compliance with MARPOL pollution prevention regulations and requirements					
.4	As instructed transfer/flood ballast water tanks to correct levels					

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
8.	**Competence: Contribute to the safe operation of equipment and machinery**				
8.1	**Safe operation of equipment, including: valves and pumps**		*Operations are carried out in accordance with established safety practices and equipment operating instructions. Communications within the operator's area of responsibility are consistently successful*		

	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Assist an officer with: Ballasting					
.2	Deballasting					
.3	Stripping tanks					
.4	Operate a lube oil purifier					
.5	Operate a fuel oil purifier/separator					
.6	Operate positive displacement pump					
.7	Operate a centrifugal pump					
.8	Start-up and put on line the emergency compressor					
.9	Start-up and put on line the emergency generator					
.10	Start-up the emergency fire pump					
.11	List other equipment you have operated:					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
8.	**Competence: Contribute to the safe operation of equipment and machinery**					
8.2	**Safe operation of equipment, including hoists and lifting equipment**			*Operations are carried out in accordance with established safety practices and equipment operating instructions. Communications within the operator's area of responsibility are consistently successful*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Operate engine-room lifting gantry under supervision					
.2	Understand hoists and lifting equipment are limited by Safe Working Load (SWL)					
.3	Identify SWL of chain blocks, shackles, slings, strops etc used for lifting					
.4	Assist with pulling main engine piston					
.5	Use stores crane for handling engine-room stores					
8.3	**Safe operation of equipment, including hatches, watertight doors, ports and related equipment**			*Operations are carried out in accordance with established safety practices and equipment operating instructions. Communications within the operator's area of responsibility are consistently successful*		
.1	Secure engine-room vents, vent shafts and fire flaps					
.2	Assist with removing crankcase doors					
.3	Understand that rubber sealing on hatches, doors and ports needs replacing periodically					
.4	Rig guard wires, chains or fences at open hatch ways, floor plates etc					
.5	Understand dangers associated with remotely operated water tight doors					
.6	Open and close water tight doors including bow, stern and shell doors					

8.4	Ability to use and understand basic crane, winch and hoist signals			*Operations are carried out in accordance with established safety practices and equipment operating instructions. Communications within the operator's area of responsibility are consistently successful*		
.1	Understand and demonstrate use of commonly used hand signals for control of lifting appliances and winches					
.2	Understand that in different cultures signals commonly used may differ					
.3	Understand the need to agree signals to be used and which person in a team is in control of lifting operation					

FUNCTION: ELECTRICAL, ELECTRONIC AND CONTROL ENGINEERING AT THE SUPPORT LEVEL (FOR RATINGS AS ABLE SEAFARER ENGINE IN A MANNED ENGINE-ROOM OR DESIGNATED TO PERFORM DUTIES IN A PERIODICALLY UNMANNED ENGINE-ROOM)

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
9.	**Competence: Safe use of electrical equipment**					
9.1	**Safe use and operation of electrical equipment, including safety precautions before commencing work or repair**			*Recognizes and reports electrical hazards and unsafe equipment. Understands safe voltages for hand held equipment*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Demonstrate a basic understanding of the risks associated with electrical equipment					
.2	Understand that the higher the voltage the greater the risk of fatal shock					
.3	Understand that high voltages may be present in low voltage equipment					
.4	Identify risks, plan and prepare before starting work					
.5	Understand what tasks require a permit to work prior to starting work					
.6	Apply knowledge of safe use of electrical equipment for use in hazardous areas					
.7	Demonstrate a knowledge of earth faults and how to avoid them					
9.2	**Safe use and operation of electrical equipment, including isolation procedures**			*Understands risks associated with high voltage equipment and on board work*		
.1	Isolate, tag and lock out where possible associated equipment when engaged in repair or maintenance work					
.2	Understand the dangers from arcs, short circuits and high voltage					
.3	Adopt correct earthing procedures for work on high voltage equipment					

9.3	**Safe use and operation of electrical equipment, including emergency procedures**			*Understands risks associated with high voltage equipment and on board work*		
.1	Demonstrate an electric shock first aid rescue					
.2	Demonstrate a knowledge of risk mitigation measures including use of insulated tools, electrical insulated gloves, rubber isolation mats etc					
9.4	**Safe use and operation of electrical equipment, including different voltages on board**			*Understands risks associated with high voltage equipment and on board work*		
.1	Assist in tracing and correcting earth faults on main and secondary electrical systems					
.2	Demonstrate recognition of factors indicating earths, e.g. broken insulation, loose equipment, signs of burning or arcing					
.3	Assist in the routine maintenance of: Circuit breakers					
.4	Tripping mechanisms					
.5	Motor starters					
.6	Lighting					
.7	Emergency storage batteries					
.8	Assist in routine testing and maintenance motors and generators					
.9	Demonstrate knowledge of hazards of working on main switch boards					
.10	Demonstrate knowledge of alternator or generator trips and their resets					
9.5	**Knowledge of the causes of electric shock and precautions to be observed to prevent shock**			*Understands risks associated with high voltage equipment and on board work*		
.1	Demonstrate an understanding of the ship's permit to work system					
.2	Apply safe working practices in using electric power operated tools and equipment					
.3	Understand the risk of shock is greater in damp or humid conditions					

FUNCTION: MAINTENANCE AND REPAIR AT THE SUPPORT LEVEL (FOR RATINGS AS ABLE SEAFARER ENGINE IN A MANNED ENGINE-ROOM OR DESIGNATED TO PERFORM DUTIES IN A PERIODICALLY UNMANNED ENGINE-ROOM)

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
10.	Competence: Contribute to shipboard maintenance and repair					
10.1	Knowledge of surface preparation techniques			*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Prepare steel work for coating using powered scaling equipment and grinders					
.2	Understand different coating sequences may be required and paint manufacturers' instructions should be checked					
.3	Understand some rust removers and degreasing agents may be corrosive and eyes and skin should be protected					
.4	Understand the difference between one pack and two pack paints					
.5	Understand two pack paints once mixed may have limited time for application					
.6	Demonstrate a knowledge of the different paints commonly used in the engine-room eg. enamel and epoxy paints					
10.2	Ability to use painting, lubrication and cleaning materials and equipment			*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
.1	Clean and maintain paintwork					
.2	Clean and maintain deck plates free of oil and grease					
.3	Apply paints using brush/roller					
.4	Apply protective coatings as directed					
.5	Understand paints may contain toxic or irritant substances					

.6	Understand need for interior and enclosed spaces to be well ventilated during and after painting					
.7	Understand solvents may give rise to flammable and potentially explosive vapours					
.8	Understand that planned maintenance schedules apply for the lubrication, overhaul and testing of lifting and mechanical equipment					
.9	Select and use correct lubricant or grease					
.10	Lubricate and grease equipment and moving parts as required					
.11	Leave areas clean and tidy on completion of work					
10.3	**Knowledge of safe disposal of waste materials**			*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
.1	Demonstrate a knowledge of the procedures for storage and disposal of paint residues, solvents and oily rags					
.2	Demonstrate a knowledge of the procedures for storage and disposal of combustion waste, waste oils and slops					
.3	Operate shredder and incinerator					
.4	Operate waste handling equipment as required					
10.4	**Ability to understand and execute routine maintenance and repair procedures**			*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
.1	Replace: Fuses					
.2	Control lamps					
.3	Lighting equipment					
.4	Assist in tracing earth faults					
.5	Check control air filters under supervision					
.6	Check control air driers and replace consumables under supervision					

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
10.	Competence: Contribute to shipboard maintenance and repair				
10.4	Ability to understand and execute routine maintenance and repair procedures (continued)		*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.7	Open up purifiers/separators for cleaning and maintenance under supervision				
.8	Re-assemble purifiers/separators under supervision				
.9	Assist with routine maintenance on the main compressor				
.10	Assist with routine maintenance on refrigeration plant				
.11	Assist with routine maintenance on fresh water generator				
.12	Assist with opening up centrifugal pump				
.13	Assist with opening up and overhaul positive displacement pump				
.14	Assist with overhaul and test valves including: Gate				
.15	Screw down non return (SDNR)				
.16	Screw lift				
.17	Relief				
.18	Two or three way cock				
.19	Shut off cock				
.20	Carry out routine maintenance of: Steering gear				

.21	Cargo winches					
.22	Cargo crane					
.23	Assist with the testing of watertight doors, ports, hatches and quick closing valves					
.24	Assist with testing of electrical emergency stops					
.25	Carry out routine maintenance of: Fire pumps					
.26	Fire flaps					
.27	Engine-room fire extinguishing system and equipment					
.28	Emergency generator					
.29	Emergency compressor					
.30	Remote stops for pumps with overboard discharges					
.31	Fuel trips					
.32	Assist in overhaul of main engine cylinder head					
.33	Assist in overhaul/replacement of main engine piston rings					
.34	Assist in replacing main engine bearing					
.35	Assist with cleaning of air side of the turbo charger					
.36	Assist with water wash exhaust side of main engine turbo charger					
.37	Assist with overhaul of auxiliary engines					
.38	Under supervision, test water gauge glasses and check that passages, cocks and valves are clear					
.39	Assist with opening up a boiler					

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
10.	**Competence: Contribute to shipboard maintenance and repair**				
10.4	**Ability to understand and execute routine maintenance and repair procedures (continued)**		*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.40	Assist with opening up: Boiler safety valve				
.41	Boiler feed check valves				
.42	List other equipment on which you have carried out routine maintenance or repairs:				
10.5	**Understand manufacturer's safety guidelines and shipboard instructions**		*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
.1	Demonstrate an understanding of the ship's permit to work system				
.2	Demonstrate an understanding of safety guidelines provided				
.3	Demonstrate an understanding of enclosed space entry routine				
.4	Demonstrate an understanding of the need for ship board familiarisation procedures				
.5	Demonstrate an understanding of what safety information may be provided in manufacturers' manuals				
.6	Demonstrate an understanding of what information may be provided in the ship's operations and maintenance manuals				

10.6	**Knowledge of the application, maintenance and use of hand and power tools and measuring instruments and machine tools**			*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
.1	List hand and power tools you have used:					
.2	List measuring equipment you have used: Liner gauge Micrometer Others:					
.3	Assist with maintenance of a starter					
.4	Assist with tracing earths					
10.7	**Knowledge of metal work**			*Maintenance activities are carried out in accordance with technical, safety and procedural specifications. Selection and use of equipment and tools is appropriate*		
.1	List below metal work you have worked on (for example, pipework, bearings, steel plate, sheet metal etc):					

FUNCTION: CONTROLLING THE OPERATION OF THE SHIP AND CARE FOR PERSONS ON BOARD AT THE SUPPORT LEVEL (FOR RATINGS AS ABLE SEAFARER ENGINE IN A MANNED ENGINE-ROOM OR DESIGNATED TO PERFORM DUTIES IN A PERIODICALLY UNMANNED ENGINE-ROOM)

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
11.	Competence: Contribute to the handling of stores					
11.1	Knowledge of procedures for safe handling, stowage and securing of stores			*Stores operations are carried out in accordance with established safety practices and equipment operating instructions. The handling of dangerous, hazardous and harmful stores complies with established safety practices. Communications within the operator's area of responsibility are consistently successful*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Assist with receiving, checking, stowage and securing ship's stores					
.2	Recognise labels that indicate stores are classified as dangerous, hazardous or harmful substances and liquids					
.3	Understand the reasons and need for segregation of such stores (solid materials and fluids)					
.4	Describe the procedure to follow in event of leakage or spillage of dangerous materials					
.5	Understand that Material Safety Data sheets provide information and advice on segregation and stowage					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
12.	Competence: Apply precautions and contribute to the prevention of pollution to the marine environment					
12.1	Knowledge of the precautions to be taken to prevent pollution of the marine environment			*Procedures designed to safeguard the marine environment are observed at all times*		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Understand that Material Safety Data sheets provide information and advice on responding to spills or leaks					
.2	Demonstrate a knowledge of the emergency shut down procedures for bunkering, deballasting, OWS, incinerator, etc.					
.3	Understand the need to segregate waste, record amounts and land ashore for disposal					
.4	Understand the need to store and dispose of oily waste according to MARPOL					
.5	Demonstrates an awareness of the need to protect the marine environment					
12.2	Knowledge of use and operation of anti-pollution equipment			*Procedures designed to safeguard the marine environment are observed at all times*		
.1	Participate in an emergency response exercise for controlling spillage of oil on board					
.2	Participate in drill for clean up of hazardous substance spillage					
.3	Demonstrate a knowledge of the procedures to collect and segregate waste and garbage					
.4	Operate garbage shredder/compactor unit and incinerator (where fitted)					
.5	Understand the need to strictly observe local regulations concerning waste water and sewage disposal					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
12.	**Competence: Apply precautions and contribute to the prevention of pollution to the marine environment**					
12.2	**Knowledge of use and operation of anti-pollution equipment (continued)**			*Procedures designed to safeguard the marine environment are observed at all times*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.6	Understand the need to strictly observe local regulations concerning flue gas emissions					
.7	Understand the need to strictly observe MARPOL regulations					
.8	Understand the need to strictly observe local regulations concerning ballast water and deballasting					
12.3	**Knowledge of approved methods for disposal of marine pollutants**			*Procedures designed to safeguard the marine environment are observed at all times*		
.1	Understand that marine pollutants must be landed ashore for safe disposal in compliance with MARPOL					
.2	Understand there are strict rules covering disposal at sea of oily water mixtures applicable to all ships					
.3	Understand there are strict rules covering disposal at sea of noxious liquid substances applicable to all ships					
.4	Understand there are strict rules covering disposal at sea of harmful substances carried in packaged form applicable to all ships					
.5	Understand there are strict rules covering pollution prevention by sewage applicable to ships					
.6	Understand the importance of strict compliance with local regulations concerning waste water and sewage disposal					
.7	Understand there are strict rules for prevention of pollution by garbage from ships at sea, applicable to all ships					
.8	Understand the need to strictly observe local regulations concerning garbage stowage and disposal ashore					

.9	Understand there are strict rules covering air pollution from ships at sea which will progressively apply to all ships					
.10	Understand the need to strictly observe local regulations concerning flue gas emissions					
.11	Understand that disposal of waste and marine pollutants is generally banned at sea and the norm is to land ashore for disposal					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
13.	Competence: Apply occupational health and safety procedures					
13.1	Working knowledge of safe working practices and personal shipboard safety, including: Electrical safety			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Demonstrate an understanding of safe working practices in using electric power operated tools and equipment					
.2	Understand risks associated with using portable electric lamps					
.3	Understand risk of shock is greater in damp or humid conditions					
.4	Demonstrate knowledge of safe use of electrical equipment for use in hazardous areas					
.5	Demonstrate use of oxygen meter for checking spaces that may be at risk					
.6	Report defective or damaged electrical equipment, connectors or cables					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
13.	Competence: Apply occupational health and safety procedures					
13.2	Working knowledge of safe working practices and personal shipboard safety, including: Lockout/tag-out			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Ensure hazardous power sources are isolated, tagged and locked and rendered inoperative before any repairs are started					
.2	Follow ship's procedure for locking and tagging to manage electrical, mechanical, pneumatic or hydraulic risks					
13.3	Working knowledge of safe working practices and personal shipboard safety, including: Mechanical safety			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
.1	Apply safe working practices in using lifting gear					
.2	Ensure correct safety guards for appliances are correctly fitted and suitable for use before start-up					
.3	Understand hazards associated with hydraulic and pneumatically operated mechanical equipment					
13.4	Working knowledge of safe working practices and personal shipboard safety, including: Permit to work systems			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
.1	Demonstrate an understanding of the ship's permit to work system					
.2	Prepare work areas in accordance with safe working practices					
.3	Stow equipment, tools and surplus materials on completion of work					
.4	Seek guidance if in any doubt concerning task or safety matters					

.5	Demonstrate an understanding of the procedure to "sign off" on completion of work					
13.5	**Working knowledge of safe working practices and personal shipboard safety, including: Working aloft**			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
.1	Demonstrate a knowledge of safe working practices when working at height					
.2	Demonstrate a knowledge of safe working practices when working overhead					
.3	Understand the risks associated with portable ladders, scaffolds and towers					
13.6	**Working knowledge of safe working practices and personal shipboard safety, including: Working in enclosed spaces**			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
.1	Demonstrate a knowledge of how to recognise an enclosed space					
.2	Seek permit to work before entry into an enclosed space (tank entry; beneath floor plates; scavenge spaces; boilers; etc)					
.3	Consistently apply safe working practices for entry into enclosed spaces					
.4	State action to take on finding casualty in an enclosed space					
13.7	**Working knowledge of safe working practices and personal shipboard safety, including: Lifting techniques and methods of preventing back injury**			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
.1	Demonstrate safe working practices in manual lifting and carrying					
.2	Demonstrate correct manual handling techniques					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
13.	**Competence: Apply occupational health and safety procedures**					
13.8	**Working knowledge of safe working practices and personal shipboard safety, including: Chemical and biohazard safety**			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
	Task/Duty	**Task Completed** Supervising Officer/ Instructor (Initials/Date)		**Advice on Areas for Improvement**	**Task Completed** Supervising Officer/ Instructor (Initials/Date)	
.1	Comply with health, hygiene and safety requirements when handling: Hazardous substances					
.2	Sewage treatment plant					
.3	Medical waste					
.4	Water ballast treatment					
.5	Follow instructions and precautions when working with cleaning fluids, paints, toxic materials etc					
.6	Understand that a biological hazard (biohazard) is a threat to human life, and that they can arise on board					
13.9	**Working knowledge of safe working practices and personal shipboard safety, including: Personal safety equipment**			*Procedures designed to safeguard personnel and the ship are observed at all times. Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times*		
.1	Seek advice to use correct personal safety equipment for the tasks in hand					
.2	Understand the need to report promptly faults or deficiencies in safety equipment					
.3	Recognise and understand meaning of prohibition, warning, mandatory and emergency safety signage					

SECTION 8 TASK SUMMARY CHART – RATINGS FORMING PART OF AN ENGINEERING WATCH

The purpose of the summary chart is to provide you, your company and your ships' chief engineers and officers with a guide and continuous check on the numbers of tasks or duties that you have completed (as listed in Section 6), and those that remain outstanding. Tick off only those tasks which you have completed.

In the charts below the tinted boxes simply indicate the start of a new group of tasks or duties.

FUNCTION – Marine Engineering at the Support Level

1. COMPETENCE – Carry out a watch routine appropriate to the duties of a rating forming part of an engine-room watch

1.1.1	1.1.2	1.1.3	**1.2.1**	1.2.2	1.2.3	1.2.4	1.2.5	1.2.6	1.2.7	1.2.8	1.2.9	1.2.10	1.2.11	1.2.12	1.2.13
1.2.14	1.2.15	1.2.16	1.2.17	1.2.18	1.2.19	1.2.20	1.2.21	1.2.22	1.2.23	1.2.24	1.2.25	1.2.26	1.2.27	1.2.28	1.2.29
1.2.30	1.2.31	1.2.32	1.2.33	1.2.34	1.2.35	1.2.36	1.2.37	1.2.38	1.2.39	1.2.40	1.2.41	1.2.42	1.2.43	1.2.44	1.2.45
1.2.46	**1.3.1**	1.3.2	1.3.3	1.3.4	1.3.5	1.3.6	1.3.7	1.3.8	1.3.9	1.3.10	1.3.11	1.3.12	1.3.13	1.3.14	1.3.15
1.3.16	1.3.17	1.3.18	1.3.19	1.3.20	1.3.21	1.3.22	1.3.23	**1.4.1**	1.4.2	1.4.3	1.4.4	1.4.5	1.4.6	**1.5.1**	1.5.2
1.6.1	1.6.2	1.6.3	1.6.4	1.6.5	1.6.6	1.6.7									

2. COMPETENCE – Maintain the correct levels and steam pressures (for keeping a boiler watch)

2.1.1	2.1.2	2.1.3	2.1.4	2.1.5	2.1.6	2.1.7	2.1.8	2.1.9	2.1.10	2.1.11	2.1.12	2.1.13	2.1.14	2.1.15

3. COMPETENCE – Operate emergency equipment and apply emergency procedures

3.1.1	3.1.2	3.1.3	3.1.4	3.1.5	3.1.6	3.1.7	3.1.8	3.1.9	3.1.10	3.1.11	3.1.12	3.1.13	3.1.14	3.1.15	3.1.16
3.1.17	3.1.18	3.1.19	3.1.20	3.1.21	3.1.22	3.1.23	**3.2.1**	3.2.2	3.2.3	3.2.4	**3.3.1**	3.3.2	3.3.3	3.3.4	3.3.5
3.3.6	3.3.7	3.3.8	3.3.9	3.3.10	3.3.11	3.3.12	3.3.13	3.3.14	3.1.15	3.3.16	3.3.17	3.3.18	3.3.19	3.3.20	

TASK SUMMARY CHART – ABLE SEAFARER ENGINE

The purpose of the summary chart is to provide you, your company and your ships' chief engineers and officers with a guide and continuous check on the numbers of tasks or duties that you have completed (as listed in Section 7), and those that remain outstanding. Tick off only those tasks which you have completed. In the charts below the tinted boxes simply indicate the start of a new group of tasks or duties.

FUNCTION – Marine Engineering at the Support Level

4. COMPETENCE – Contribute to a safe engineering watch

4.1.1	4.1.2	4.1.3	4.1.4	**4.2.1**	4.2.2	4.2.3	4.2.4	4.2.5	4.2.6	4.2.7	4.2.8	4.2.9	**4.3.1**	4.3.2	4.3.3
4.3.4	4.3.5	4.3.6	4.3.7	4.3.8	4.3.9										

5. COMPETENCE – Contribute to the monitoring and controlling of an engine-room watch

5.1.1	5.1.2	5.1.3	5.1.4	5.1.5	5.1.6	5.1.7	5.1.8	5.1.9	5.1.10	5.1.11	5.1.12	5.1.13	5.1.14	5.1.15	5.1.16
5.1.17	5.1.18	5.1.19	5.1.20	5.1.21	5.1.22	5.1.23	5.1.24	5.1.25	5.1.26	5.1.27	5.1.28	5.1.29	5.1.30	5.1.31	**5.2.1**
5.2.2	5.2.3	5.2.4	5.2.5	5.2.6	5.2.7	5.2.8	5.2.9	5.2.10	5.2.11	5.2.12	5.2.13	5.2.14	5.2.15	5.2.16	5.2.17
5.2.18	5.2.19	5.2.20	5.2.21	5.2.22	5.2.23										

6. COMPETENCE – Contribute to fuelling and oil transfer operations

6.1.1	6.1.2	6.1.3	6.1.4	6.1.5	**6.2.1**	6.2.2	6.2.3	**6.3.1**	6.3.2	6.3.3	6.3.4	**6.4.1**	6.4.2	**6.5.1**	6.5.2
6.5.3	6.5.4	6.5.5													

7. COMPETENCE – Contribute to bilge and ballast operations

7.1.1	7.1.2	7.1.3	7.1.4	7.1.5	**7.2.1**	7.2.2	7.2.3	7.2.4

8. COMPETENCE – Contribute to the safe operation of equipment and machinery

8.1.1	8.1.2	8.1.3	8.1.4	8.1.5	8.1.6	8.1.7	8.1.8	8.1.9	8.1.10	8.1.11	**8.2.1**	8.2.2	8.2.3	8.2.4	8.2.5

8.3.1	8.3.2	8.3.3	8.3.4	8.3.5	8.3.6	**8.4.1**	8.4.2	8.4.3

FUNCTION – Electrical, Electronic and Control Engineering at the Support Level

9. COMPETENCE – Safe use of electrical equipment

9.1.1	9.1.2	9.1.3	9.1.4	9.1.5	9.1.6	9.1.7	**9.2.1**	9.2.2	9.2.3	**9.3.1**	9.3.2	**9.4.1**	9.4.2	9.4.3	9.4.4

9.4.5	9.4.6	9.4.7	9.4.8	9.4.9	9.4.10	**9.5.1**	9.5.2	9.5.3

FUNCTION – Maintenance and Repair at the Support Level

10. COMPETENCE – Contribute to shipboard maintenance and repair

10.1.1	10.1.2	10.1.3	10.1.4	10.1.5	10.1.6	**10.2.1**	10.2.2	10.2.3	10.2.4	10.2.5	10.2.6	10.2.7	10.2.8	10.2.9	10.2.10
10.2.11	**10.3.1**	10.3.2	10.3.3	10.3.4	**10.4.1**	10.4.2	10.4.3	10.4.4	10.4.5	10.4.6	10.4.7	10.4.8	10.4.9	10.4.10	10.4.11
10.4.12	10.4.13	10.4.14	10.4.15	10.4.16	10.4.17	10.4.18	10.4.19	10.4.20	10.4.21	10.4.22	10.4.23	10.4.24	10.4.25	10.4.26	10.4.27
10.4.28	10.4.29	10.4.30	10.4.31	10.4.32	10.4.33	10.4.34	10.4.35	10.4.36	10.4.37	10.4.38	10.4.39	10.4.40	10.4.41	10.4.42	**10.5.1**
10.5.2	10.5.3	10.5.4	10.5.5	10.5.6	**10.6.1**	10.6.2	10.6.3	10.6.4	**10.7.1**						

FUNCTION – Controlling the Operation of the Ship and Care for Persons on Board at the Support Level

11. COMPETENCE – Contribute to the handling of stores

11.1.1	11.1.2	11.1.3	11.1.4	11.1.5

12. COMPETENCE – Apply precautions and contribute to the prevention of pollution of the marine environment

12.1.1	12.1.2	12.1.3	12.1.4	12.1.5	**12.2.1**	12.2.2	12.2.3	12.2.4	12.2.5	12.2.6	12.2.7	12.2.8	**12.3.1**	12.3.2	12.3.3
12.3.4	12.3.5	12.3.6	12.3.7	12.3.8	12.3.9	12.3.10	12.3.11								

13. COMPETENCE – Apply occupational health and safety procedures

13.1.1	13.1.2	13.1.3	13.1.4	13.1.5	13.1.6	**13.2.1**	13.2.2	**13.3.1**	13.3.2	13.3.3	**13.4.1**	13.4.2	13.4.3	13.4.4	13.4.5
13.5.1	13.5.2	13.5.3	**13.6.1**	13.6.2	13.6.3	13.6.4	**13.7.1**	13.7.2	**13.8.1**	13.8.2	13.8.3	13.8.4	13.8.5	13.8.6	**13.9.1**
13.9.2	13.9.3														